

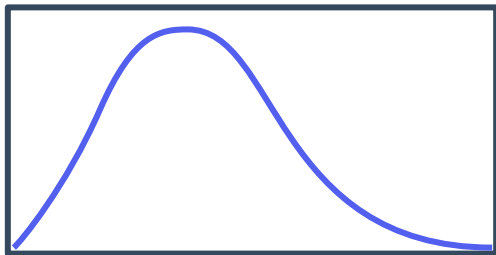
Normal distributions



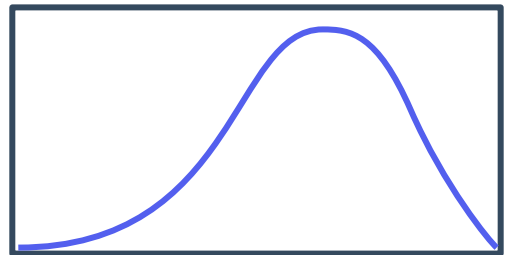
- 1 State whether the following statements are true of the normal distribution curve:
- a There are the same number of scores above and below the mean.
 - b The curve is symmetrical.
 - c The fewest scores lie around the mean.
 - d The curve is asymmetrical.
 - e The spread of the normal distribution changes depending on the mean.
 - f A higher mean will result in a skewed curve.
 - g The mean, median and mode all have the same value.
 - h The spread of the normal distribution changes depending on the standard deviation.

- 2 State whether the following figures are normally distributed:

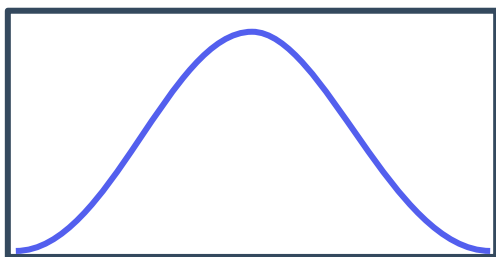
a



b



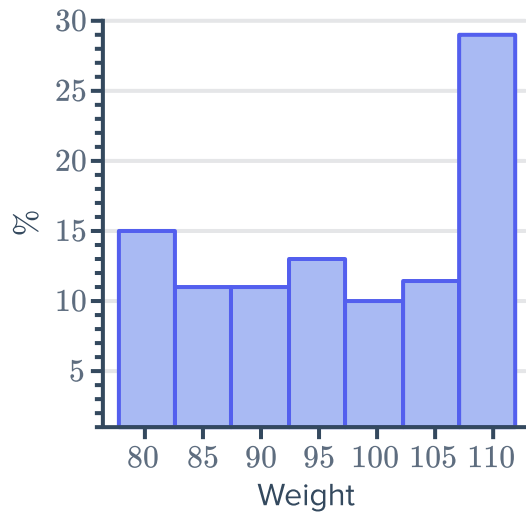
c



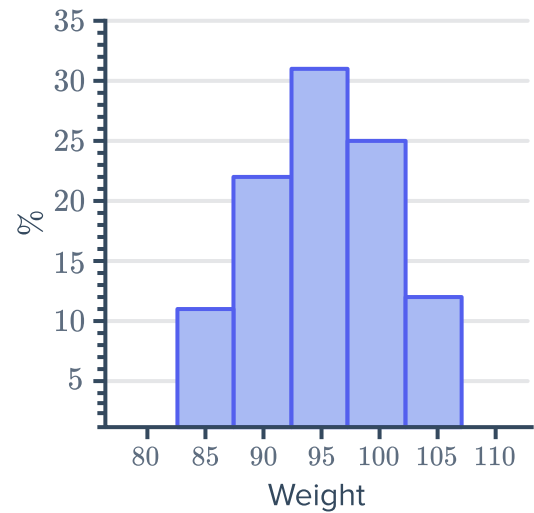
3 State whether the following histograms are approximately normally distributed:



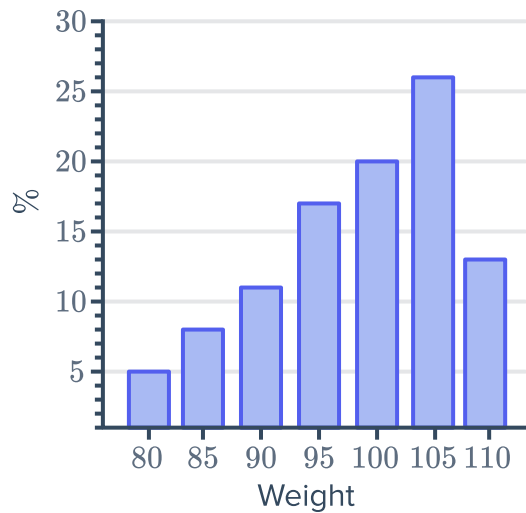
a



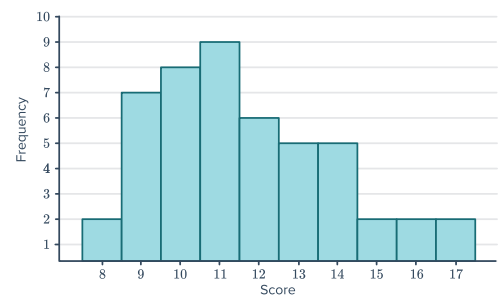
b



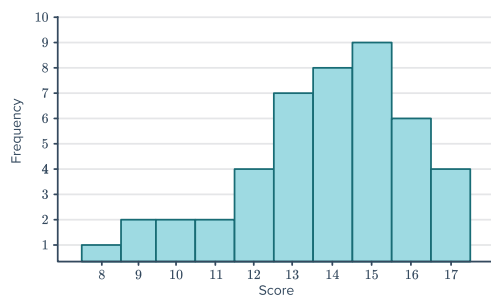
c



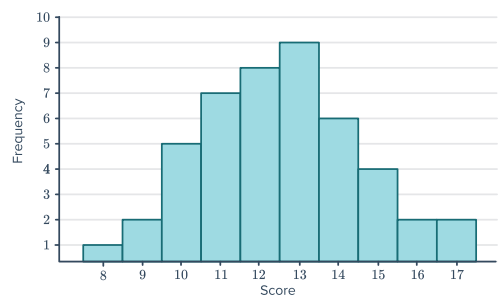
d



e



f



- 4 State whether the following boxplots could be from data which is approximately normally distributed:



a



b



c



- 5 Given a normal distribution, state the percentage of people that lie:

a Above the mean.

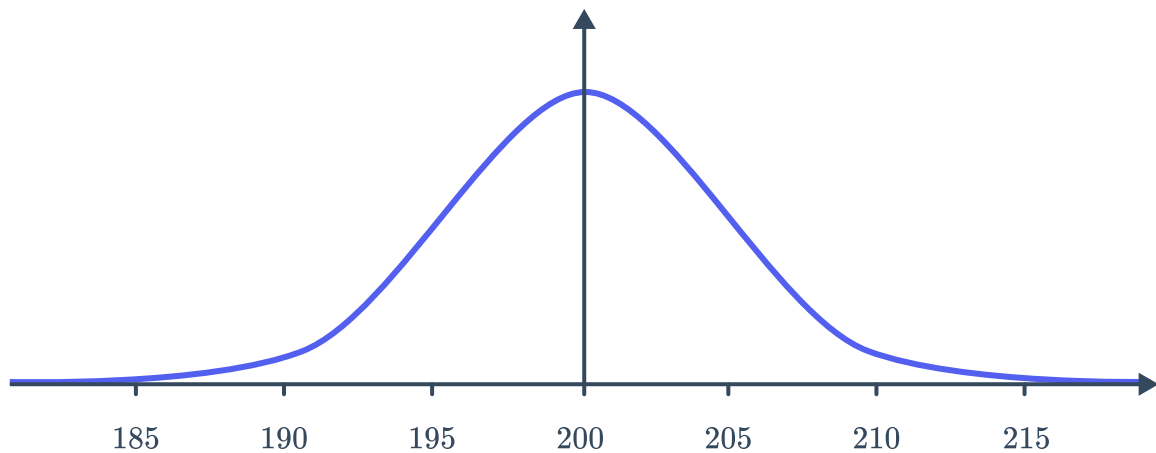
b Below the mean.

6 Consider the given normal distribution:



- a State the mean.
- c State the mode.

b State the median.

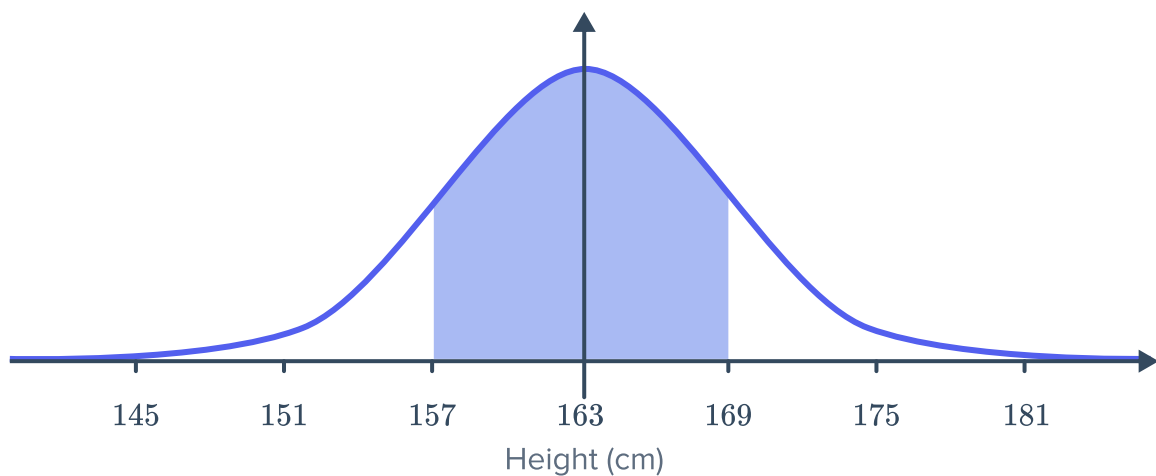


7 The mean of a set of scores, denoted by μ is 51 and the standard deviation, denoted by σ is 16. Find the value of the following:

- a $\mu - \sigma$
- b $\mu + 2\sigma$
- c $\mu + 3\sigma$
- d $\mu - 2\sigma$
- e $\mu + 0.5\sigma$
- f $\mu - \frac{2\sigma}{3}$

8 A data set has a mean of 80 and a standard deviation of 3. Find the score that is 3 standard deviations above the mean.

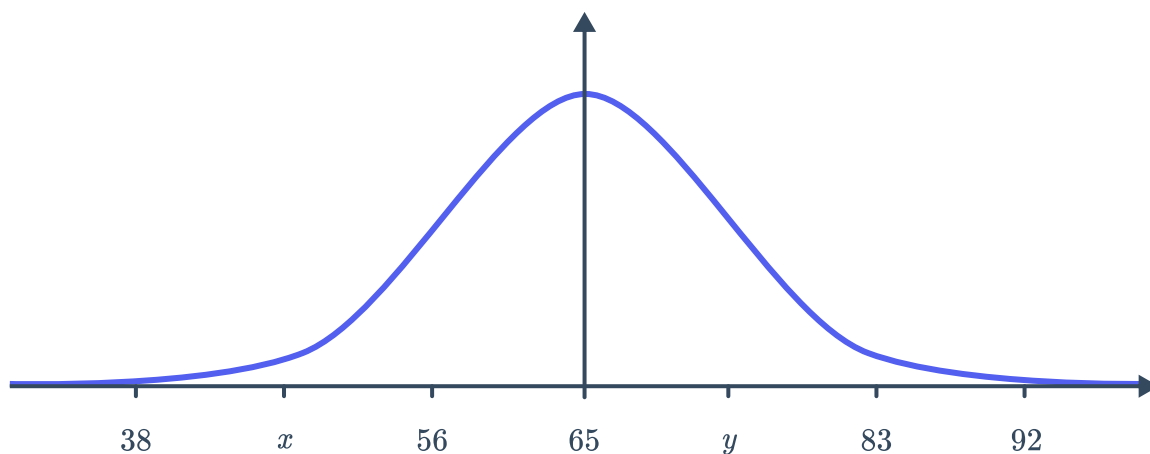
9 The following graph shows the distribution of females' heights in a population:



- a If 34% of females lie between 157 cm and 163 cm, find the percentage of females that lie between 163 cm and 169 cm.
- b State the value of 1 standard deviation.



- 10 The results from an exam were approximately normally distributed. The mean score was 65, with a standard deviation of 9. The points marked on the horizontal axis are separated by 1 standard deviation:



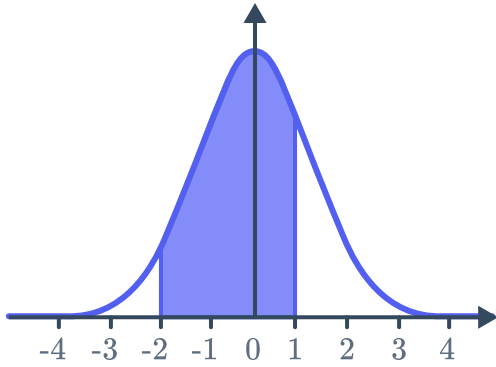
- a State the value of x on the graph. b State the value of y on the graph.
- 11 Assume the mass of sumo wrestlers is normally distributed, with a mean mass of 158 kg and a standard deviation of 10 kg.
- a State the percentage of wrestler that would weigh more than 158 kg.
- b How far below the mean is a sumo wrestler who weighs 148 kg.
- 12 In a male population, the mean height was 175 cm, with a standard deviation of 7 cm.
- a State the height of Bob who is 4 standard deviations above the mean.
- b State the height of John who is 2 standard deviations below the mean.
- 13 A sample of professional basketball players is normally distributed and gives the mean height as 199 cm with a standard deviation of 10 cm. State the height of a basketball player who is:
- a 3 standard deviations above the mean b 1.5 standard deviations below the mean



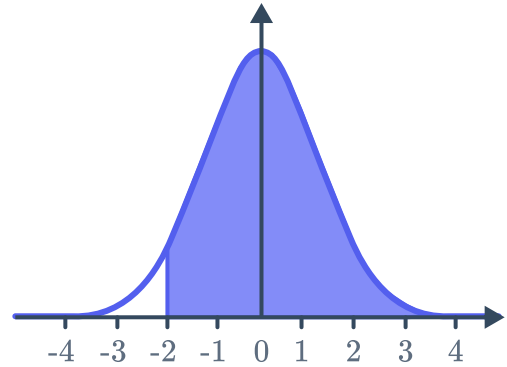
The empirical rule

14 In the following normal distributions, each unit on the horizontal axis indicates 1 standard deviation. Find the approximate percentage of scores that lie in the shaded region:

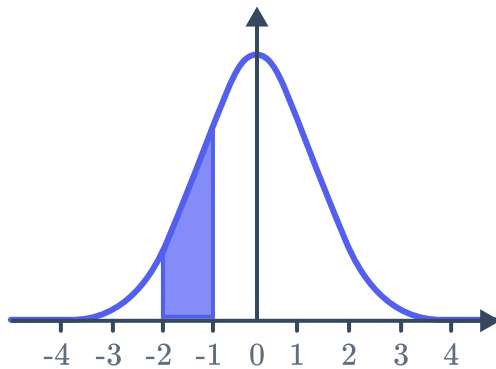
a



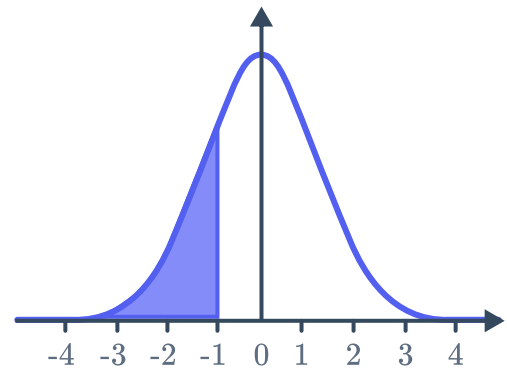
b



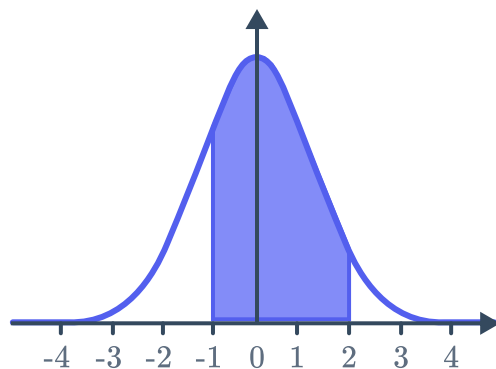
c



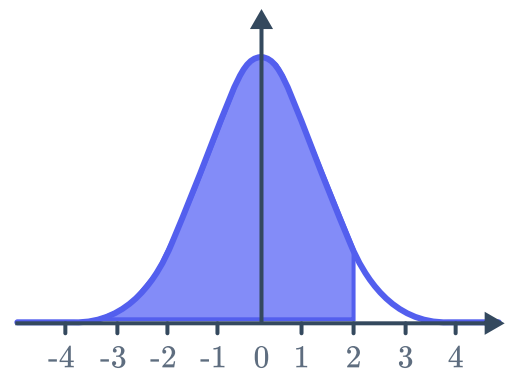
d



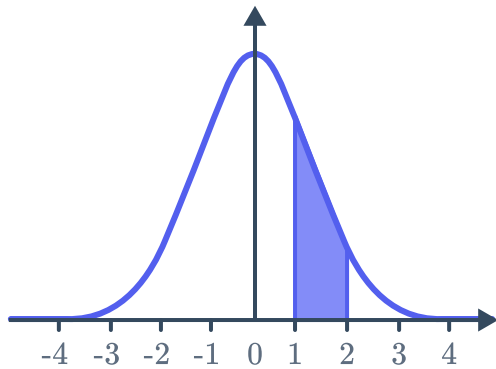
e



f



g



- 15** In a normal distribution, state the approximate percentage of the scores that lie within:
- a** 1 standard deviation of the mean.
 - b** 2 standard deviations of the mean.
 - c** 3 standard deviations of the mean.
- 16** In a normal distribution, state the approximate percentage of scores that lie between the mean and:
- a** 1 standard deviation above the mean.
 - b** 2 standard deviations below the mean.
 - c** 3 standard deviations below the mean.
- 17** In a normal distribution, state the approximate percentage of scores that lie between:
- a** 1 standard deviation above and 2 standard deviations below the mean.
 - b** 1 standard deviation below and 3 standard deviations above the mean.
 - c** 2 standard deviations below and 3 standard deviations above the mean.
- 18** In a normal distribution, the percentages of scores that lie within x standard deviation(s) of the mean are given below. Find the value of x for each percentage:
- a** Approximately 68% of scores lie within x standard deviation(s) of the mean.
 - b** Approximately 95% of scores lie within x standard deviation(s) of the mean.
 - c** Approximately 99.7% of scores lie within x standard deviation(s) of the mean.

19 In a normal distribution, the percentages of scores that lie between the mean and x standard deviation(s) above the mean are given below. Find the value of x for each percentage.

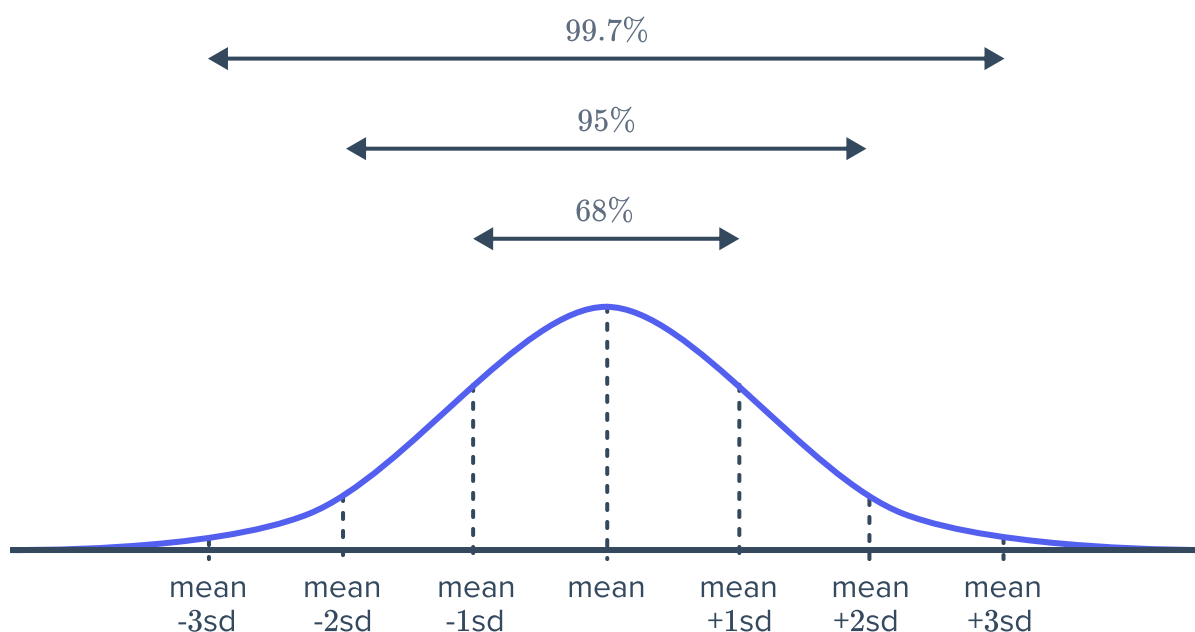


- a Approximately 34% of scores lie between the mean and x standard deviation(s) above the mean.
- b Approximately 47.5% of scores lie between the mean and x standard deviation(s) above the mean.
- c Approximately 49.85% of scores lie between the mean and x standard deviation(s) above the mean.

20 The grades in a test are approximately normally distributed. The mean mark is 60 with a standard deviation of 2. The approximate percentages of results that lie symmetrically about the mean are given below. Find the two scores that each of the following percentages lie between:

- a Approximately 68%.
- b Approximately 95%.
- c Approximately 99.7%.

21 The figure shows the approximate percentage of scores lying within various standard deviations from the mean of a normal distribution:



The heights of 600 boys are found to approximately follow such a distribution, with a mean height of 145 cm and a standard deviation of 20 cm. Find the number of boys with heights between:

- a 125 cm and 165 cm.
- b 105 cm and 185 cm .
- c 85 cm and 205 cm .
- d 145 cm and 165 cm .
- e 165 cm and 185 cm .



- 28** The marks in a class were approximately normally distributed. If the mean was 47 with a standard deviation of 6, Find the percentage of students who achieved:
- a** A mark above the average.
 - b** A mark above 41.
 - c** A mark above 53.
 - d** A mark below 35.
 - e** A mark above 29.
 - f** A mark between 29 and 47.
 - g** A mark between 53 and 59.
- 29** The times that a class of students spent talking or texting on their phones on a particular weekend is approximately normally distributed with mean time 173 minutes and standard deviation 4 minutes. Find the percentage of students that used their phones for between 165 and 181 minutes on the weekend.
- 30** The height of sunflowers is approximately normally distributed, with a mean height of 1.6 m and a standard deviation of 5 cm.
- a** Find the percentage of sunflowers between 1.5 m and 1.75 m tall.
 - b** Find the percentage of sunflowers between 1.55 m and 1.75 m tall.
 - c** If there are 3000 sunflowers, find the number of sunflowers taller than 1.5 m.
- 31** The exams scores of students are approximately normally distributed with a mean score of 63 and a standard deviation of 8. Use the empirical rule to find:
- a** The approximate percentage of students who scored between 39 and 79.
 - b** The approximate number of students who passed if there are 450 students in the class and if the passing score is 39.
- 32** The operating times of phone batteries are approximately normally distributed with mean 34 hours and a standard deviation of 4 hours.
- a** Find the percentage of batteries that last between 22 and 42 hours.
 - b** Find the percentage of batteries that last between 30 hours and 42 hours.
 - c** Any battery that lasts less than 22 hours is deemed faulty. If a company manufactured 51 000 batteries, find the approximate number of batteries they would be able to sell.
- 33** The number of biscuits in a box is approximately normally distributed with mean 30 and standard deviation of 3.
- a** Approximately 81.5% of the scores lie between 2 standard deviations below and x standard deviation(s) above the mean. Find the value of x .
 - b** Find the range of the numbers of biscuits in 81.5% of the boxes.



34 The weights of an adult harp seals are approximately normally distributed with mean 144 kg and standard deviation of 6 kg.

- a** Approximately 83.85% of adult harp seals lie between 1 standard deviation below and x standard deviation(s) above the mean. Find the value of x .
- b** Hence, find the range of the weight of 83.85% adult males, in kilograms.

35 The times for runners to complete a 100 m race is approximately normally distributed with mean 14 seconds and standard deviation of 1.9 seconds.

- a** Approximately 97.35% of people lie between 3 standard deviations below and x standard deviation(s) above the mean. Find the value of x .
- b** Hence, find the range of time in seconds for which 97.35% of runners completed the race. Round your answer to one decimal place.